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Citation in BibTeX format

ICIMMI 2023: International Conference on Information Management & Machine Intelligence
November 23 - 25, 2023
Jaipur, India

RESEARCH-ARTICLE

Smart Agriculture for Sustainable Growth: Realtime Assessment of Macronutrient Analysis using IoT Sensors and Parallel Hybrid CNN_LSTM for Crop and Fertilizer Optimization

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ICIMMI '23: Proceedings of the 5th International Conference on Information Management & Machine Intelligence (November 2023)
https://doi.org/10.1145/3647444.3647922
ISBN: 9798400709418

Smart Agriculture for Sustainable Growth: Real-time Assessment of Macronutrient Analysis using IoT Sensors and Parallel-Hybrid CNN-LSTM for Crop and Fertilizer Optimization.

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ABSTRACT

Food security and sustainable growth depend on smart agriculture. Real-time macronutrient analysis using the Internet of Things (IoT) and artificial intelligence (AI) is essential for crop and fertilizer optimization because soil quality is crucial to plant growth and productivity. This study proposes a novel parallel-hybrid CNN-LSTM model with Bayesian optimization for smart agriculture soil nutrient management. Modern farming relies on resource efficiency and waste reduction. IoT technology helps farmers integrate smart farming practices for real-time monitoring and data-driven decision-making. With IoT sensors, soil data like macronutrient levels, soil temperature, and pH values are continuously collected to assess soil health. The hybrid model better captures spatial patterns and temporal dependencies in sequential data, matching soil nutrient and environmental data. Based on real-time soil nutrient data, the proposed approach identified suitable crops and fertilizers with 99.12% accuracy after rigorous experimentation and evaluation. The parallel-hybrid CNN-LSTM model and Bayesian optimization make precise and timely decisions for optimal crop growth and resource management in smart agriculture, as shown by this high accuracy rate. This strategy aims to boost agricultural productivity, sustainability, and resource efficiency to create a more resilient and eco-friendly agricultural ecosystem.

KEYWORDS

Soil nutrients, Macro-nutrient, Internet of Things (IoT), Sensors, NPK, Artificial Intelligence

ACM Reference Format:
Saurabh Bhattacharya and Dr. Manju Pandey. 2023. Smart Agriculture for Sustainable Growth: Real-time Assessment of Macronutrient Analysis using IoT Sensors and Parallel-Hybrid CNN-LSTM for Crop and Fertilizer Optimization.. In *International Conference on Information Management & Machine Intelligence (ICIMMI 2023), November 23–25, 2023, Jaipur, India.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3647444.3647922

1 INTRODUCTION

Ten billion people are expected by 2050. The U.N. estimates food production must rise 70% to meet demand. Conventional nutrient management hasn't changed despite many agricultural changes [1]. To determine crop nutrient needs, farmers must make assumptions. This method wastes fertilizers and pesticides, degrading the environment. Previous crop management and weather can affect field nutrient status. Soil conditions affect field nutrients. Farmers must know what nutrients to add to soil. Knowing soil nutrients increases yield. A micronutrient has copper, nickel, manganese, iron, zinc, and Molybdenum. It's impossible to grow biomass without nutrients. Plants grow with water, carbon dioxide, and nutrients. Plants mainly use macro and micronutrients. The main plant nutrients are N, P, and K. These nutrients are vital to plant tissue. N and P are plant tissue nutrients, making them essential. They contain proteins and nucleic acids. K is needed to regulate enzymes and osmosis in plants, but organic compounds lack it [2], [3].

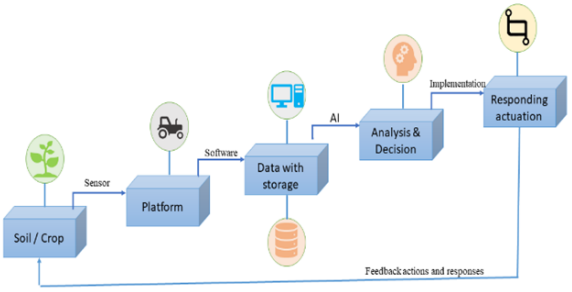

Figure 1: IoT based Precision Agriculture

Soil quality analysis boosts productivity. Assess its condition using soil pH, temperature, and nutrients. However, India has only 3887 soil testing labs, making soil quality analysis difficult for farmers and cultivators. Farmers improve crop quality and yields with soil nutrients and organic compounds. Excess nitrogen fertilizer can cause leaching and denitrification. To maximize soil potential, farmers must improve soil quality. Many industries use AI, IIoT, and machine learning and many data mining tools, techniques, and procedures are available. Smart agriculture grows sustainably with wireless communications. Precision agriculture is growing on small and large farms. Technology helps farmers collect more accurate data. Figure-1 depicts the use of IoT in precision farming, where IoT sensors and devices collect field data to make informed decisions and improve the efficiency of the crop. The data collected by the sensors is then retrieved using various methods for further processing [4]–[6].

Agricultural AI enables precision farming helps to analyze massive amounts of data and provides insights beyond human analysis.

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ICIMMI 2023, November 23–25, 2023, Jaipur, India
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ACM ISBN 979-8-4007-0941-8/23/11
https://doi.org/10.1145/3647444.3647922

Table 1: Major Related work wrt IoT

Author	Methodology	Algorithm used	Result
Andrianto et al.[8]	IoT-based service system	Decision tree	Acc. = 90%
Thorat et al.[9]	TPF-CNN	CNN	Acc. = 95%
Garg et al.[10]	Mobile vision	CNN	Acc. = 92%
Khan et al.[11]	IoT, machine learning	SVM	Acc. = 93%
Lavanya et al.[12]	IoT, low-cost sensors	Fuzzy logic	Acc. = 85%

Table 2: Major related works wrt precision agriculture

Author	Method	Finding	Drawback
Shadi AlZu'bi et al.[13]	Sensor-based	Improved crop yield and Water requirement of soil.	Image-based analysis, needs time to process. Costly Sensors and Infrastructure
Vu Bang Pham et al. [14]	AWD Tube	Water level of the field.	Limited to rice crop only.
K. R. Gsangaya et al. [15]	Sensor-based	temperature, humidity, light intensity, and soil moisture of soil.	Macro-Nutrient values are missing, able to manage the smaller area.
Akanksha Gupta et al. [16]	Machine Learning Based	Predicting the crop yield.	Real-time data collection is missing.
Marcos A.N. Coutinho et al. [17]	Vis-NIR spectroscopy.	Soil sample preparation	Only for phosphorus and potassium. Limited to total Nitrogen.
A.A Maroli et al. [18]	Sensor-based.	Water management and distribution system	Limited to soil nutrient and water pH values.
Md. Azahar Ali et al. [19]	Electrochemical sensing technology.	Soil quality assessment	Theoretical approach.

AI helps farmers choose the best fertilizer. The CNN-LSTM parallel hybrid architecture is ideal for smart agriculture. This helps the system process sequential and spatial data. The parallel hybrid CNN-LSTM model in smart agriculture works better with Bayesian optimization. The system recommends fertilizer and crop combinations more accurately using soil analysis. This study analyzes macronutrients in real-time using IoT sensors and a parallel hybrid CNN-LSTM model with Bayesian optimization. The study optimizes fertilizer and crop recommendations using this analysis [7].

1.1 Our contribution

Improves smart agriculture crop and fertilizer recommendations. Using real-time soil nutrient data, the novel parallel-hybrid CNN-LSTM model with Bayesian optimizer finds suitable crops and fertilizers with 99.12% accuracy.

2 RELATED WORK

Technology has transformed agriculture by improving resource utilization and crop production. Soil micronutrient monitoring is a promising IoT application in agriculture. Micronutrients in soil affect plant health and yield. However, soil nutrient surveillance methods are usually lengthy and laborious. IoT allows the agricultural sector to monitor these levels in real-time, improving the efficiency of various agrarian operations. Major related works wrt IoT is discussed in table 1.

This review uses multiple methods to assess soil micronutrient analysis. The study uses machine learning, spectroscopy, and electrochemical sensing. By comparing methods, one can better understand the micro-nutrient market. It also highlights areas for further research, such as developing an IoT-based micronutrient analysis system. This review emphasizes the importance of a comprehensive micronutrient monitoring strategy in agriculture. The need for more efficient soil nutrient analysis inspired this design. Major related works wrt precision agriculture is discussed in table 2.

Technological advances have enabled modern agriculture to use cutting-edge tools and methods. This opens up new ways to address field issues like diseases, nutritional deficiencies, and pests and diseases. Table 3 shows that researchers have used various methods to identify these issues more accurately and effectively, improving crop quality and yield. This review critically evaluates studies on using various technologies, algorithms, and methods to address these challenges.

3 IOT BASED PROTOTYPE

The NPK sensor (JXBS-3001-NPK-RS)) is made up of stainless steel and a high-density epoxy resin body corrosion- and rust-resistant. It protects the sensor from harsh environments. PLCs can connect to the device via 485 interfaces. It can connect to a microcontroller via 485 chips. It can also be connected to a computer via USB to 485 converters. The power interface works with 12–24 volts. An analog

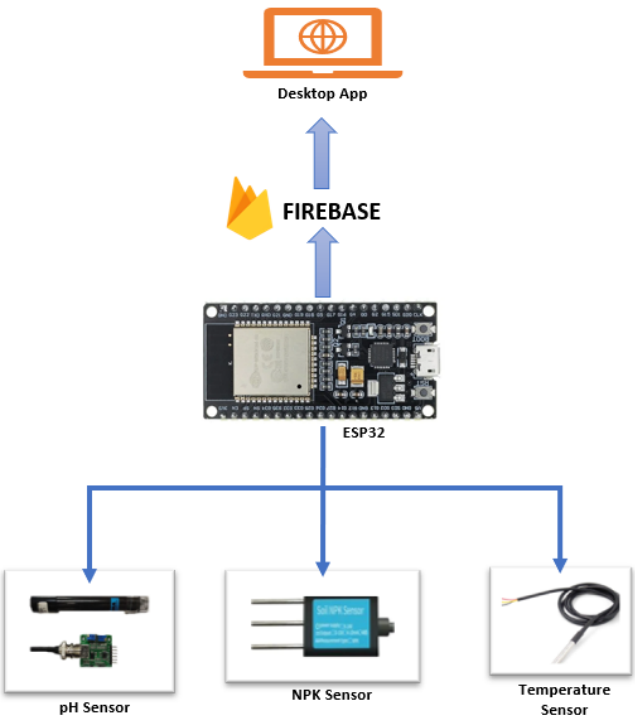

Figure 2: Proposed Prototype

pH sensor electrode with an amplifier circuit measures hydrogen ion activity in a solution. It determines acidity or alkalinity. It tests water and soil quality. Farmers and home gardeners use soil pH meters. Plants need the right pH to thrive.

The ideal plant pH is 5.5–7.5. Some plants survive outside this range. The DS18B20 Waterproof soil Temperature Sensor Probe sensors are accurate to 12 bits. A unique 64-bit I.D. distinguishes it from others. These work well with 3.3-5.0V systems. Figure-2 shows a novel proposed soil nutrient analysis data acquisition unit that this project will develop. It is rugged and suitable for harsh environments. The device's components are water-resistant to withstand harsh conditions. It should withstand drops and animal bites. Its durable construction and 12v A.C. supply make it portable. It should also work without affecting sensors. Various sensors collect field data, which is wirelessly transferred to an LCD/computer. The proposed system uses ESP32, a single-chip Bluetooth, and the company's 40 nm ultra-low-power Wi-Fi solution. Precision agriculture and industrial devices benefit from it. Sensors collect and monitor soil nutrient parameters and send the data to an LCD/monitor, desktop application, and sensorData.csv file. The proposed system can transmit data via serial cable if local Wi-Fi is unavailable as shown in figure 2.

4 METHODOLOGY

The field of deep learning is expected to revolutionize various sectors. This paper explores the necessary steps in implementing deep learning in addressing real-world challenges. The initial step in

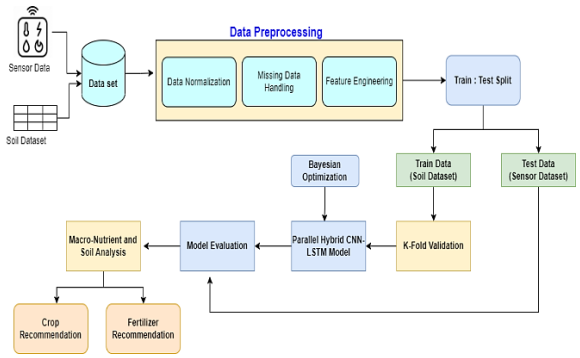

Figure 3: Proposed Methodology

implementing deep learning is to gather data. Before DL can start, the collected data must undergo preprocessing. This can involve removing outliers and cleaning the data. Once the data has been processed, the next step is to create new features in the data preprocessing steps. After creating the features, the next step is to choose a DL learning model. The chosen model needs to be trained using the collected data. This process involves feeding it the data and adjusting its parameters to see how it performs. After the training phase has been completed, the model must be subjected to an evaluation on a sensor dataset. Figure 3 the proposed methodology.

4.1 Dataset

The dataset collection process for soil health in India is an essential step in developing a machine-learning model that can accurately predict soil health conditions. The process is divided into two phases: collecting a training dataset and collecting a sensor dataset for testing. The training dataset trains the various DL models, including the proposed model. This dataset is collected from the government of India's soil health card dataset. The sensor dataset is used to test the performance of the DL model. This dataset can be collected from various IoT sensors in real-time scenarios.

4.2 Pre-Processing

Normalization: When analysing the data collected by an IoT sensor, the various features must have the same scale to prevent them from dominating the analysis. Z-Score normalization is used as a normalization technique.

- Missing Data Handling: Due to various factors, such as communication issues or sensor failure, the data collected by an IoT sensor may have missing values. To avoid skewed analysis, it is important to handle such data. Here average mean values are used for replacing the values.
- Feature Engineering: When raw IoT sensor data is analyzed, it may contain irrelevant information and a lot of noise. Feature engineering helps identify and transform the various features into useful tools for analysis.

4.3 Algorithm Used

4.3.1 Multi-Layer Perceptron (MLP).. A is an artificial neural network that consists of multiple layers of nodes or neurons. These layers include an input layer, one or more hidden layers, and an

output layer. MLPs are frequently employed in the field of machine learning to perform tasks such as classification and regression. Every node within the network is linked to nodes in neighbouring layers, and these connections possess weights that are modified during training to acquire knowledge of patterns and relationships within the data. MLPs are renowned for their capacity to capture intricate patterns and non-linear correlations, rendering them adaptable for diverse applications.

$$z = \sigma\left(\sum_{i=1}^{N} w_i \cdot x_i + b\right) \tag{1}$$

where, z = "output of the neuron", σ = "activation function".

4.3.2 Random Forest. Random Forest is a technique in ensemble learning that merges the predictions of numerous decision trees to enhance overall accuracy and resilience. Every decision tree is built using a randomly selected subset of the training data and features. The ultimate prediction is made by combining the predictions of each tree. Random Forests are highly efficient for both classification and regression tasks, and they demonstrate exceptional performance in managing extensive datasets with a high number of dimensions. They exhibit robustness against overfitting and serve as an effective tool for enhancing the predictive performance of machine learning models.

$$P\,(y = j|x,\ \theta) = \frac{1}{T}\sum_{t=1}^{T} P_t(y = j|x) \tag{2}$$

where Pt(y = j|x) = "probability of class j for a given input x in the tth decision tree", θ = "parameters of the Random Forest Classifier".

4.4 Proposed Method - Parallel-Hybrid CNN_LSTM classifier

The proposed novel Parallel-Hybrid CNN-LSTM (PHCL) is developed to conduct soil nutrient analysis, recommend crop types, and optimize fertilizer application. This method handles the "spatiotemporal" characteristics of soil data effectively and makes accurate recommendations for crop selection and fertilizer application by combining two powerful neural network architectures, CNN and LSTM, in a parallel manner. The sequential data represents a time series of soil nutrient measurements obtained at varying intervals.

Each data point in the sequence contains information about the levels of nutrients found in various parts of your field located at a specific spatial location. The CNN was altered to function on a 1D sequence in this scenario. To accomplish this, 1D convolutional layers are utilized to capture spatial patterns throughout the sequence of nutrient measurements. The sequential nutrient data are processed so the CNN can extract relevant spatial features. LSTM layers effectively capture the temporal dependency that is present in sequential data. Modelling patterns that change with time lends itself well to the use of LSTMs. It understands how nutrient levels change over varying time intervals and considers the historical context of the data. Algorithm-1 is the proposed model's algorithm.

Both the CNN and LSTM components perform their functions simultaneously. The CNN processes the sequential nutrient data to capture spatial features, while the LSTM captures temporal patterns and dependencies in the nutrient measurements. After that, the outputs from both components are combined and sent onward through subsequent layers to undergo additional analysis. When making forecasts, it is necessary to use the combined results from the parallel and hybrid components.

The model recommends the crop and fertilizer that will produce the best results based on the sequential soil nutrient data that has been analyzed. The LSTM component helps capture patterns in nutrient variations over time, which can be extremely helpful for gaining an understanding of the most suitable crops to cultivate during various times of the year or under varying environmental conditions. In optimizing fertilizer use, the model makes recommendations regarding the kind and amount of fertilizer that should be applied, considering the patterns of historical nutrient levels. The understanding of temporal dependencies provided by the LSTM helps make fertilization recommendations that align with the soil's ever-evolving nutrient requirements. Figure-4 represents the model summary.

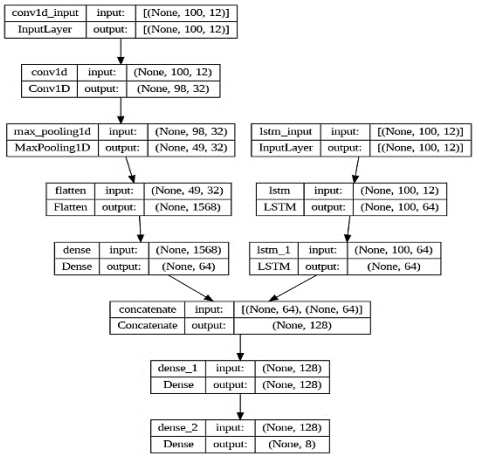

Figure 4: Model summary of Parallel-Hybrid CNN_LSTM

Let sequential soil nutrient data as a time series with measurements at different time steps and spatial locations. Time steps: $t = 1,2,3...T$ (where T= total no. of steps), Spatial locations $s=1,2,3...S$ (where s= total no. of spatial locations), Nutrient feature: $f=1,2,3...F$ (where F= total no. of nutrient features). Then mathematically CNN and LSTM are represented as:

Applying 1D convolutions over the time and spatial dimensions, the output of the CNN is represented as C, where C[t,s,c]= "output of the cth conv filter at the timestep t and spatial location s", then C represented as eq. 3

$$C\,[t, s, c] = \sum_{i}\sum_{j}\sum_{f} W_c\,[i, j, f]\,.\,X\,[t+1,\ s+j, f] + b_c \tag{3}$$

Where W_c= "filter weights of the cth conv filter ", b_c= "bias term".

Again, the LSTM layer with hidden states H, to calculate the hidden state at each time step using LSTM equations depicted from eq. 4-9:

$$i_t = \sigma\,(W_{xi}.X\,[t] + W_{hi}.H\,[t-1] + b_i) \tag{4}$$

$$f_t = \sigma\left(W_{xf}.X\,[t] + W_{hf}.H\,[t-1] + b_f\right) \tag{5}$$

$$o_t = \sigma\,(W_{xo}.X\,[t] + W_{ho}.H\,[t-1] + b_o) \tag{6}$$

$$g_t = tanh\left(W_{xg}.X\left[t\right] + W_{hg}.H\left[t-1\right] + b_g\right) \quad (7)$$

$$c_t = f_t.c_{t-1} + i_t.g_t) \quad (8)$$

$$H\left[t\right] = o_t.\tanh\left(c_t\right) \quad (9)$$

where, i_t, f_t, o_t, g_t = "input, forget, output and candidate cell gate at time step t", W_{xi}, W_{hi}, W_{hf}, W_{xo}, W_{ho}, W_{xf}, W_{xg}, W_{hg} = "Weight matrices", b_i, b_f, b_o, b_g = "bias term for different gates".

4.5 Bayesian Optimization

Using Bayesian optimization techniques helps improve operations' efficiency by identifying optimal hyperparameters and minimizing noise. Here in Smart Agriculture, IoT sensors and a parallel-hybrid CNN-LSTM model analyze the data for crop recommendations and optimize fertilizer recommendations. The objective function is a composite measure of how well the system performs, such as how accurate it is when forecasting crop yield or optimizing fertilizer applications. It is represented as f(x) with the representation of the various configurations or hyperparameters that can be used to improve the efficiency of operations.

Surrogate Model: The process of Bayesian optimization involves creating a surrogate function, also referred to as a surrogate model, that approximates the actual objective function of f(x). The most common choice for this type of model is the Gaussian Process. The "surrogate model" $m(x)$ is represented as a Gaussian Process with a mean function $\mu(x)$ and a covariance function $k(x, x^{'})$. The observed data $D = (x_i, y_i)$, where $y_i = f(x_i)$ is observed value at point x_i, the predictive distribution Gaussian process is depicted in eq. 10.

$$m\left(x\right) = \mu\left(x\right) + k(x)^T\left(K + \sigma^2 I\right)^{-1}\left(y - \mu\right) \quad (10)$$

where, $k(x)$= "vector of covariance between x and observed data", K = "covariance matrix of the observed data points", σ^2= "noise variance", I = "identity matrix".

Initial Data: The initial data points are collected by evaluating the objective function of the selected points. The surrogate model is then constructed using these points.

Acquisition Function: The objective function of a given project is then computed using a surrogate model. It considers exploring and exploiting the target area to arrive at the next step. The acquisition function $\alpha(x)$ guides the selection of the next point. The Expected Improvement (E.I.) acquisition function is defined as in eq. 11.

$$\alpha_{EI}\left(x\right) = E\left[\max\left(0, f_{min} - f\left(x\right)\right)\right] \quad (11)$$

Point Selection: The acquisition function determines the optimal point in the exploration-exploitation trade-off. The point x_{next} is selected by maximizing the acquisition function depicted as in eq. 12.

$$x_{next} = argmax_x \alpha\left(x\right) \quad (12)$$

5 RESULTS AND OUTPUT

5.1 Evaluation parameters

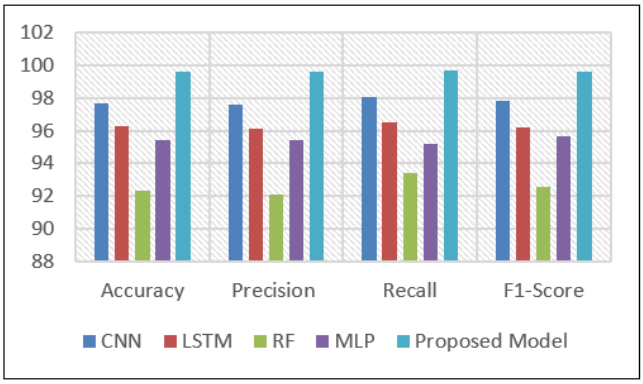

Figure 5: Comparison of various models

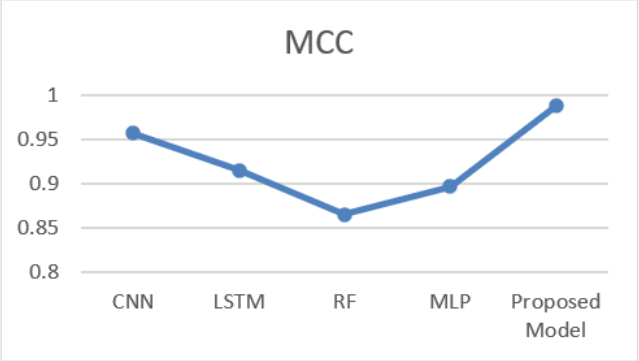

Figure 6: MCC comparison

Table 3: Evaluation Parameters comparison of various classifiers

Classifiers	Accuracy	Precision	Recall	F1-Score	MCC
CNN	97.68	97.6	98.07	97.8	0.957
LSTM	96.24	96.12	96.5	96.23	0.915
RF	92.33	92.08	93.42	92.55	0.865
MLP	95.45	95.43	95.21	95.64	0.897
Proposed Model	**99.12**	**99.6**	**99.7**	**98.6**	**0.988**

5.2 Training and Validation accuracy and Loss graph – Proposed Model

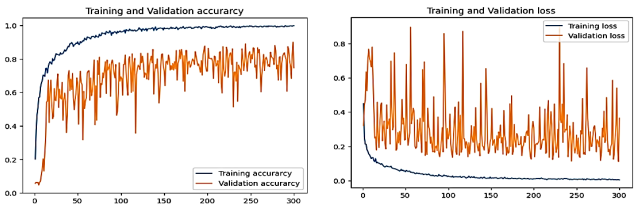

Figure 7: Training and validation - accuracy and loss graph

5.3 Comparison with major existing work

Table 4: Comparison with existing work

Model	Accuracy
IDCSO-WLSTM[20]	92.68
GNB[11]	97
LSTM[21]	86.3
Proposed model	**99.12**

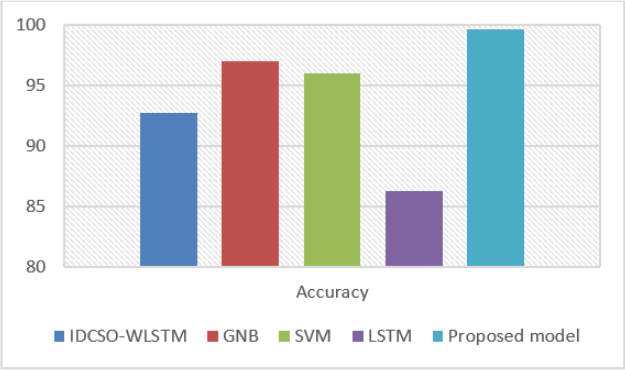

Figure 8: Comparison with existing work

Table 3 and fig. 5,6 show the interesting results of classifier evaluation for smart agriculture crop and fertilizer recommendations. Critical metrics like "accuracy, precision, recall, F1-score, and the Matthews Correlation Coefficient" were used to evaluate the classifier's prediction abilities. The "CNN" had 97.68% accuracy, 97.6% precision, 98.07% recall, 97.8% F1-score, and 0.957 MCC. Similarly, the "LSTM" model performed well with 96.24% accuracy, 96.12% precision, 96.5% recall, 96.23% F1-score, and 0.915 MCC. The "Random Forest" classifier had 92.33% accuracy, 92.08% precision, 93.42% recall and 92.55% F1-score, and 0.865 MCC. The "MLP" had 95.45% accuracy, 95.43% precision, 95.21% recall and 95.64% F1-score, and 0.897 MCC. Fig.7 compares training and validation accuracy and loss.

The innovative parallel-hybrid CNN-LSTM architecture with Bayesian optimization, the Proposed Model, performed best. With 99.12% accuracy, 99.6% precision, and 99.7% recall, this model excelled. Although the F1-score was lower at 98.6%, the MCC remained high at 0.988, demonstrating the model's robustness. As shown in table 4, fig. 8, the Proposed Model outperformed the IDCSO-WLSTM (92.68%), GNB (97%), SVM (96%), and LSTM (86.3%) with 99.12% accuracy. The parallel-hybrid CNN-LSTM model and Bayesian optimization pioneered smart agriculture by providing precise crop and fertilizer recommendations.

6 CONCLUSION AND FUTURE SCOPE

In the pursuit of promoting sustainable growth and ensuring food security, this study embarked on a pioneering journey in the domain of smart agriculture. By harnessing the potential of IoT sensors and AI A novel approach was proposed for real-time macronutrient analysis to optimize crop yield and fertilizer recommendations. The significance of soil quality in supporting plant growth cannot be overstated, and the fusion of IoT technology and AI-driven insights is poised to revolutionize modern farming practices. The heart of this research lies in the parallel-hybrid CNN-LSTM model, fortified by Bayesian optimization, which showcases a formidable amalgamation of CNN and LSTM networks. This architecture's prowess in capturing spatial patterns and temporal dependencies aligns seamlessly with the complexities inherent in soil nutrient and environmental data. The impressive results obtained, with an accuracy of 99.12%, speak volumes about the model's precision in identifying optimal crops and fertilizers based on real-time soil nutrient data. The implications of these findings are far-reaching. With the proposed approach, farmers and growers are equipped with a potent tool for informed decision-making, resulting in optimal resource utilization, minimal wastage, and increased agricultural productivity. By fostering a deeper understanding of soil health and nutrient dynamics, this strategy not only bolsters crop growth but also upholds the principles of sustainability and resource efficiency, thereby contributing to a resilient and environmentally conscious agricultural paradigm. The horizons of this research beckon towards several promising directions. Expanding the research to encompass various geographical regions and diverse crops would enhance the model's applicability and robustness.

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